Efficient Brain MRI Classification with Sub-1M Parameter Models and Multi-Scale Consistency-Guided Inference

Anonymous Authors

***Abstract*—Deep learning has transformed medical image analysis, becoming central to many clinical applications. However, real-world deployment remains challenging due to computational demands, latency constraints, and privacy concerns associated with cloud-based processing. Despite these limitations, state-of-the-art models increasingly favor larger architectures and datasets, reinforcing a scale-driven paradigm that is often impractical for resource-constrained systems. This study systematically evaluates deep learning models with fewer than one million parameters on brain MRI classification and histological glioma grading, focusing on edge efficiency and cross-dataset generalization. On saturated classification benchmarks, compact models achieve diagnostic accuracy statistically comparable to architectures an order of magnitude larger; the top-performing compact model reaches 98.29% accuracy at 0.36 GFLOPs, executing six times faster than ResNet-50 with less than a one-percentage-point difference in accuracy. However, a performance gap remains on more challenging cross-dataset tasks, with errors concentrated in low-confidence predictions sensitive to input perturbations. The most efficient architecture, which uses parallel atrous convolution branches to capture multi-scale features, is particularly vulnerable to scale-sensitive inputs. To address this limitation, we propose MCG-TTA, an architecture-informed inference strategy designed to improve robustness while limiting computational overhead. MCG-TTA improves slice-level F1 by an average of 1.43 percentage points (and up to 1.83 percentage points) at approximately 1.9× the baseline latency, while retaining a model size of 0.91 million parameters. These findings suggest that, for resource-constrained clinical applications, uncertainty-aware inference can recover residual accuracy more efficiently than increasing model complexity.**

***Index Terms*—Magnetic resonance imaging (MRI), brain tumor classification, compact neural networks, edge computing, test-time augmentation, glioma grading, uncertainty estimation.**

I. Introduction

DEEP neural architectures have driven major advances across fundamental computer vision tasks, ranging from large-scale visual categorization [8], [19] to dense spatial reasoning and semantic localization [10], [37]. Early progress was established by Convolutional Neural Networks (CNNs) [15], [18], which introduced translation equivariance and hierarchical spatial representations as core architectural inductive biases. To address optimization degradation in deeper networks, residual learning [10] and compound scaling formulations [30] subsequently enabled systematic scaling across network depth, width, and input resolution. Concurrently, vision research has expanded toward Vision Transformers (ViTs) [6], [33], masked autoencoding [34], and self-supervised foundation representations [1], where relaxing strict local inductive biases facilitates dynamic long-range contextual dependencies. However, these architectural paradigms primarily scale performance by increasing model capacity to tens or hundreds of millions of parameters, coupled with intensive pretraining regimes on high-throughput compute clusters [8], [32] (detailed across dominant paradigms in Section II).

In clinical radiology, deep learning has demonstrated diagnostic efficacy competitive with subspecialized clinicians across diverse imaging modalities [7], [28]. Within neuro-oncology, non-invasive Magnetic Resonance Imaging (MRI) serves as the primary diagnostic modality for the screening, differential triage, and histologic grading of intracranial neoplasms [2], [26], [38]. Clinical decision-making requires precise differentiation among distinct pathological entities (primarily diffuse gliomas, extra-axial meningiomas, sellar pituitary adenomas, and non-tumorous tissue), which are associated with distinct treatment pathways ranging from urgent surgical resection and adjuvant therapy to conservative surveillance [26], [40]. Radiologically, this differential triage requires careful evaluation of complex tissue characteristics: distinguishing infiltrative margins from well-circumscribed borders, assessing microvascular permeability and contrast enhancement, and differentiating non-enhancing peritumoral edema from necrotic cores across multi-parametric acquisitions, including T1-weighted, contrast-enhanced T1ce, T2-weighted, and FLAIR sequences [2], [28], [38]. However, manual multi-planar slice inspection across axial, coronal, and sagittal volumes places substantial cognitive demands on neuroradiologists, contributing to diagnostic variability and triage delays [7], [40]. To automate intracranial lesion classification, clinical research has largely adapted established, overparameterized computer vision backbones [7], [39], [40], which require substantial computational and memory resources during deployment.

Despite strong empirical performance, a translational gap remains between retrospective offline benchmark accuracy and practical point-of-care deployment [21], [25]. In clinical environments, deployment rarely relies on centralized, high-throughput cloud servers. Transmitting high-resolution volumetric neuroimaging to external cloud infrastructures introduces privacy and governance constraints, including compliance with patient confidentiality frameworks such as HIPAA in the United States and GDPR in the European Union [21]. Furthermore, cloud-dependent pipelines can be vulnerable to

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network latency, variable bandwidth, and connectivity disruptions, which present challenges during acute clinical scenarios, emergency neuro-triage, and intra-operative navigation [21], [25]. Consequently, executing inference directly on local edge hardware, such as diagnostic workstations, scanner consoles, and mobile intra-operative units, provides a practical pathway for decentralized clinical deployment [9], [20], [21].

Edge deployment imposes specific physical constraints: bounded inference latency, constrained memory footprints, restricted thermal budgets, and reliance on unaccelerated host CPUs or low-power embedded processors [9], [25]. To accommodate resource-constrained environments, lightweight architectures have leveraged efficient operators such as depthwise separable convolutions [11], [27] and inverted residual bottlenecks [22] (detailed in Section II). Nevertheless, the majority of established lightweight networks operate within mobile-scale budgets of 2 to 20 million parameters. In contrast, the sub-1M parameter regime ($< 1,000,000$ parameters) remains underexplored in neuro-oncological imaging, particularly under the constraints of small-to-medium institutional datasets [3], [16]. Furthermore, uniformly scaling down standard architectures via width or depth multipliers can destabilize gradient flow and restrict feature diversity in small-to-medium clinical cohorts [16], [19].

This investigation addresses the lack of systematic empirical benchmarks and architectural insights for deep learning models operating under a strict sub-1M parameter budget in neuro-oncology. We examine how architectural topology, specifically the allocation of parameters across multi-scale branches without structural redundancy, influences feature representation when model capacity is tightly constrained [16]. While multi-path atrous (dilated) depthwise convolutions provide an effective mechanism for expanding effective receptive fields without parameter inflation [5], [16], their spatial response characteristics and stability under clinical distribution shifts require systematic evaluation.

To address these questions, this work presents a systematic empirical evaluation of deep learning models operating under a strict sub-1M parameter budget across representative neuro-oncological tasks. Our findings highlight two primary phenomena:

1) **In-Domain Saturation**: On canonical diagnostic benchmarks, structurally optimized compact models match the classification performance of overparameterized backbones an order of magnitude larger while achieving substantially lower latency on unaccelerated edge processors. On constrained clinical cohorts, excessive parameter capacity yields diminishing returns.
2) **Cross-Dataset Scale Sensitivity**: Under cross-cohort distribution shifts involving variable slice thickness and lesion scales, compact networks with parallel atrous branches exhibit increased prediction variance, with uncertainty concentrating along ambiguous tumor boundaries due to discrete spatial dilation grids.

Because multi-model ensembles and larger backbones incur prohibitive memory and latency overheads on edge hardware [9], [16], recovering cross-domain robustness requires efficient inference-time mechanisms that operate within existing model constraints. To this end, we introduce **Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA)**. Rather than executing unconditional multi-view ensembles across all inputs [4], [35], MCG-TTA monitors predictive uncertainty and selectively deploys targeted scale perturbations that align with the internal receptive field offsets of the atrous backbone. Full multi-view augmentation is triggered only when scale disagreement is detected on ambiguous inputs, maintaining baseline single-pass speed on confident samples and requiring zero additional learned parameters.

Experimental evaluations demonstrate that uncertainty-guided test-time calibration stabilizes compact representations under distribution shifts while preserving patient-level diagnostic sensitivity. These results indicate that architecture-aligned inference calibration offers a practical, hardware-friendly strategy for edge-deployed medical imaging systems.

In summary, the principal contributions of this work are:

- **Sub-1M Architectural Benchmarking and Edge Profiling**: We provide a systematic empirical evaluation of sub-1M parameter vision models on neuro-oncological imaging, characterizing trade-offs across model capacity, single-thread edge CPU latency, throughput, and generalization against mobile and overparameterized baselines, supported by patient-level diagnostic evaluation.
- **Characterization of Saturation and Discrete Scale Dynamics**: We analyze the performance saturation of compact models on in-domain benchmarks and identify the scale-dependent sensitivity of compact atrous convolutions under cross-institutional domain shifts.
- **Architecture-Aligned Test-Time Calibration**: We introduce MCG-TTA, a parameter-free, uncertainty-gated test-time augmentation framework that selectively stabilizes scale-sensitive predictions with minimal computational overhead, preserving edge real-time throughput.

II. Related Work

Deep neural networks have advanced medical computer vision, yet their clinical adoption in neuro-oncology remains constrained by the memory, compute, and latency profiles of prevailing models. In this section, we review relevant literature across dominant architectural paradigms, examining their mathematical foundations, applications in brain MRI, and practical limitations on edge hardware.

A. Deep Learning for Neuro-Oncology and the Burden of Scale

As convolutional networks grew deeper, optimization was hindered by vanishing and exploding gradients, as well as degradation where accuracy saturates and abruptly drops. He *et al.* [10] addressed this degradation with Residual Networks (ResNet), introducing identity shortcut connections that allow error signals to bypass parameterized convolutions without attenuation. In neuro-oncology, deep residual architectures, particularly ResNet-18, ResNet-50, and ResNet-101, have been widely used across tumor detection, multi-class classification, and histological grading [7], [28], [38], [39]. Surveys by Litjens *et al.* [7] and Lundervold and Lundervold [38],

alongside studies such as Nayak *et al.* [39], demonstrate that deep CNNs achieve near-saturated accuracies (97.5–100%) on multi-class brain MRI benchmarks [39], a pattern also reproduced on public datasets like the Nickparvar cohort [26].

Beyond purely convolutional designs, the Vision Transformer (ViT) [6] divides images into coordinate patches and applies global self-attention [33]. While ViTs offer substantial representational capacity when pre-trained on large external datasets, their lack of translational equivariance and localized inductive biases makes them susceptible to overfitting when trained from scratch on small-to-medium medical cohorts [19]. Furthermore, the quadratic computational complexity of self-attention with respect to sequence length increases latency and memory demands during high-resolution volumetric MRI processing [6], [46]. To combine transformer design principles with convolutional efficiency, Liu *et al.* [24] developed ConvNeXt, modernizing ResNets with 7×7 depthwise kernels, inverted bottlenecks, and LayerNorm. ConvNeXt V2 [34] incorporated masked autoencoders and global response normalization to further stabilize representation learning.

Despite competitive benchmark metrics, deep networks carry practical limitations for clinical workflows. Standard convolutional backbones such as ResNet-50 [10] and VGG-16 [29] comprise tens to hundreds of millions of parameters, demanding substantial on-device storage and computational bandwidth. As documented by Mittal *et al.* [25] and noted in neuro-oncology surveys [40], diagnostic workstations and intra-operative systems often rely on general-purpose central processing units (CPUs) or embedded chips without high-end GPU acceleration. On CPU platforms, deep residual networks face noticeable latency overheads: benchmarks by Bianco *et al.* [43] show ResNet-50 achieving under 19 frames per second, with single-image latency frequently reaching 50 to 80 ms [25], [44]. For volumetric examinations containing dozens or hundreds of slices across multi-parametric pulse sequences (T1, T1ce, T2, FLAIR) [2], [23], [45], this processing time accumulates to several seconds per scan. Furthermore, multi-branch residual additions create high memory access costs and off-chip traffic, leading to cache contention that widens the gap between theoretical FLOPs and actual execution time [25].

B. Lightweight Architectures and Multi-Scale Representations

To provide computationally lightweight models suitable for edge execution, several architectural strategies have emerged in computer vision:

Factorized and Inverted Bottleneck Convolutions: Depthwise separable convolutions were popularized by Chollet [13] in Xception and generalized by Howard *et al.* [11] in MobileNet. This decomposition decouples spatial filtering from channel mixing: a depthwise convolution applies a single spatial kernel per channel, followed by a 1×1 pointwise convolution that linearly combines features across channels. For 3×3 kernels, this factorization achieves an 8- to 9-fold reduction in arithmetic operations with modest impact on representational capacity. Sandler *et al.* [27] introduced MobileNetV2, utilizing inverted residual blocks with linear bottlenecks. Unlike standard residual blocks that compress feature channels before spatial convolution, MobileNetV2 expands inputs to a higher-dimensional space, performs spatial depthwise filtering, and projects features back to a lower-dimensional subspace to avoid non-linear activation collapse. MobileNetV3 [14] extended this approach using hardware-aware neural architecture search (NAS) [36], hard-swish activations, and Squeeze-and-Excitation (SE) channel recalibration blocks [12].

Compound Network Scaling: The EfficientNet family [30] systematically balances accuracy and computational cost through compound scaling. Instead of adjusting depth, channel width, or image resolution in isolation, Tan and Le demonstrated that these dimensions are interdependent, scaling them jointly via a constant compound coefficient, with EfficientNet-B0 establishing an efficient baseline for mobile applications [30]. To address memory locality and training speed on modern processors, EfficientNetV2 [31] adopted Fused-MBConv blocks, replacing early depthwise layers with standard 3×3 convolutions.

Multi-Scale Feature Representation and Dilated Convolutions: Distinguishing intracranial tumor boundaries, perifocal edema, and necrotic tissue requires multi-scale spatial context [2], [23], [39]. Standard CNNs capture multi-scale context by downsampling feature maps, which discards high-frequency boundary information. In contrast, atrous (dilated) convolutions [5] expand the effective receptive field without spatial downsampling or parameter inflation by inserting spaces between kernel weights. Chen *et al.* [5] applied this via Atrous Spatial Pyramid Pooling (ASPP) in DeepLab, using parallel atrous branches with distinct dilation rates to capture multi-scale context simultaneously. Kyrkou and Theocharides [17] employed multi-scale atrous fusion in EmergencyNet for drone image analysis. Building on these principles, Kyrkou [16] proposed LiteSpeed-Net (LS-Net), showing that multi-path atrous convolutions provide an effective inductive bias for compact vision models under a 1M parameter budget: by splitting channels across parallel depthwise branches with dilation rates of 1, 2, and 3, LS-Net covers wide receptive fields while maintaining low latency. However, because dilated kernels sample spatial grids at discrete intervals without interpolation [5], [16], their predictions can become sensitive to input scale variations and cross-scanner differences, such as varying slice thickness or zoom levels.

C. Test-Time Augmentation and Aleatoric Uncertainty Estimation

In safety-critical clinical applications, communicating predictive uncertainty alongside diagnostic predictions is beneficial [7], [41]. In brain MRI, ambiguity often arises from anatomical variations, scanner noise, and poorly circumscribed lesion margins, termed data-dependent aleatoric uncertainty [4], [35]. Ayhan and Berens [4] and Wang *et al.* [35] demonstrated that Test-Time Augmentation (TTA) provides a practical sampling framework to estimate aleatoric uncertainty without Bayesian retraining. By evaluating stochastically transformed views of an input slice and aggregating predictions, TTA smooths the decision boundary and produces calibrated probability estimates [4], [35], [42].

Nonetheless, conventional TTA introduces computational overhead: running an unselective multi-view ensemble (typically 6 to 8 forward passes per image) scales inference latency linearly [42], substantially reducing the efficiency gains achieved by lightweight models [35], [42]. Furthermore, Shanmugam *et al.* [42] observed that unconditionally augmenting high-confidence samples can alter canonical orientations and inadvertently degrade accuracy.

D. Open Challenges in Edge Neuro-Oncological Computing

These observations highlight several unresolved challenges in edge-oriented neuro-oncology:

1) *Limited Benchmarking Under 1M Parameters*: Medical vision studies predominantly focus either on overparameterized models ($>$ 20M parameters) or mobile-scale models (2M to 20M parameters). Systematic benchmarks for the sub-1M parameter regime ($< 1,000,000$ parameters) remain limited across in-domain classification and out-of-domain clinical grading.
2) *Discrete Sampling Sensitivity Under Domain Shift*: While parallel atrous convolutions provide high efficiency on saturated in-domain datasets, their discrete sampling mechanics can induce sensitivity under cross-dataset scale shifts, where prediction errors concentrate along lesion boundaries.
3) *Inference Overhead in Conventional TTA*: Standard test-time augmentation pipelines execute fixed, brute-force ensembles that multiply latency. Few approaches use predictive uncertainty to selectively trigger augmentation while matching transformations to the internal dilation structure of compact backbones.

To address these challenges, this study evaluates sub-1M parameter vision models on brain MRI benchmarks and introduces Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA), recovering out-of-domain diagnostic accuracy without adding learned parameters.

III. Methodology

Motivated by the compute and latency constraints of edge devices [9], [20], [21], and the observation that compact models relying on parallel dilated convolutions can exhibit sensitivity to input scale variations [5], [16], this work introduces Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA). The underlying hypothesis is that scale-induced ambiguity in compact multi-dilation architectures concentrates in low-confidence predictions, and that these predictions can be stabilized at inference time by probing the model with multi-scale views aligned with its internal dilation structure, requiring no additional learned parameters.

In this section, we formulate the image classification problem, analyze the spatial scale sensitivity of compact atrous convolutions, present the algorithmic stages of MCG-TTA, derive an analytical latency model, and define the patient-level volumetric diagnostic aggregation.

A. Problem Formulation and Base Network Architecture

Let $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$ denote a 2D brain MRI slice, where H, W represent spatial dimensions and C denotes channel dimensionality. A deep neural network f_θ parameterized by learned weights θ maps the input slice to a vector of logits $\mathbf{z} = f_\theta(\mathbf{I}) \in \mathbb{R}^K$ over K mutually exclusive diagnostic categories.

The predicted class posterior probabilities are computed via the softmax operator:

$$\mathbf{p} = \sigma\left(f_\theta(\mathbf{I})\right) \in \Delta^{K-1}, \quad \text{where } p_c = \frac{\exp(z_c)}{\sum_{j=1}^{K}\exp(z_j)} \tag{1}$$

The model's predictive confidence is quantified by the maximum posterior probability:

$$p_{\max} = \max_{c\in\{1,\ldots,K\}} p_c \tag{2}$$

with initial predicted class $c_0 = \arg\max_c p_c$.

To support execution on resource-constrained hardware, we adopt a compact multi-scale convolutional architecture as the base backbone (specifically LiteSpeed-Net [16]). As illustrated in Fig. 1, the backbone incorporates multi-path atrous (dilated) depthwise convolutions (MPAC blocks [16]) to expand the effective receptive field without parameter inflation, allowing the network to capture both local tissue textures and broader anatomical context.

B. Spatial Scale Sensitivity in Compact Atrous Convolutions

While atrous convolutions expand the receptive field without downsampling spatial resolution [5], their discrete sampling mechanics introduce structural sensitivity under input distribution shifts.

An atrous convolution with dilation rate $d > 1$ samples input features on a sparse regular lattice with stride d. In deep, overparameterized architectures, the resulting sparse sampling artifacts, commonly termed the *gridding effect* [5], are mitigated by dozens of subsequent parameterized layers that continuously blend feature maps across deep residual stages. In contrast, compact vision models designed for low-power edge environments [16] possess limited parameter depth, restricting their capacity to interpolate across discrete spatial sampling gaps.

When deployed across clinical environments with differing acquisition settings (such as variations in slice thickness, scanner geometry, or lesion dimensions), pathological margins undergo spatial scale shifts relative to the training distribution. Consequently, fixed dilated sampling points can misalign with fine anatomical boundaries, such as thin infiltrative borders or micro-structural margins [5], [16].

Based on these observations, we formulate two design principles:

1) *Uncertainty Concentration*: Spatial scale sensitivity manifests primarily in predictions exhibiting high predictive uncertainty ($p_{\max} < \tau$, where τ denotes an empirical confidence threshold).
2) *Architecture-Informed Scale Probing*: Because this sensitivity stems from discrete spatial sampling intervals,

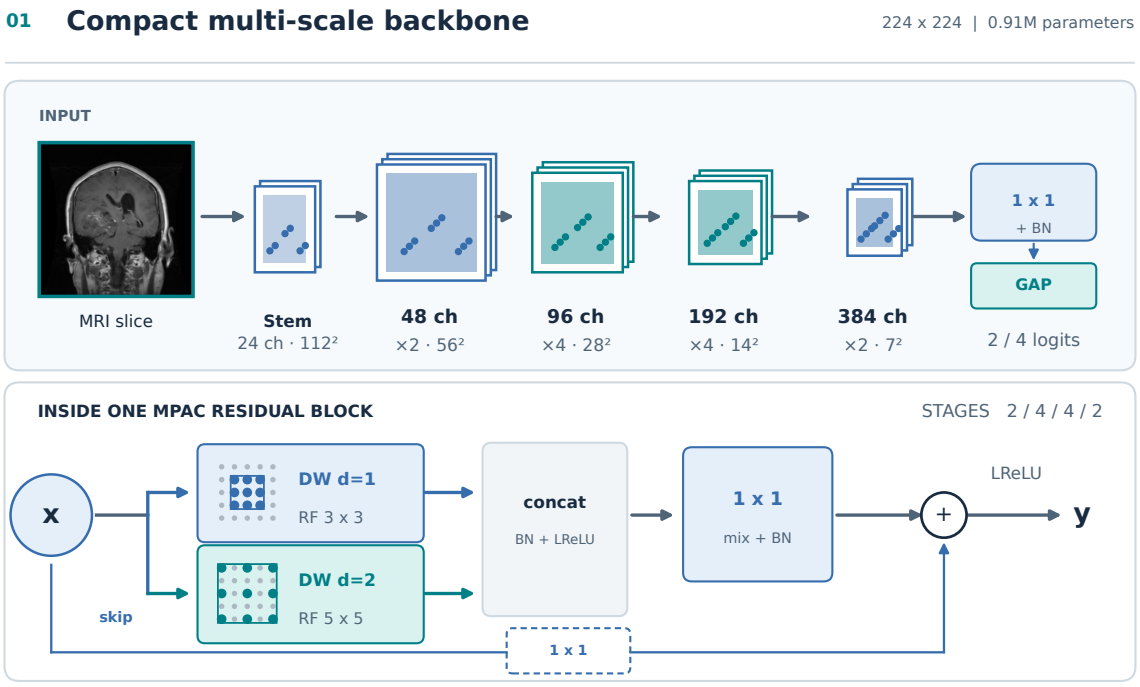


Fig. 1. Architecture of the compact LiteSpeed-Net (LS-Net) backbone and MPAC residual block.

uncertain predictions can be probed and stabilized at test time by applying targeted scale perturbations that mirror the internal dilation offsets of the architecture, without altering the model's learned parameters.

C. Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA)

To improve inference robustness while strictly bounding computational overhead, we formulate Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA). Instead of applying test-time augmentation unconditionally across all inputs, which scales inference latency linearly with the number of views [42], MCG-TTA dynamically routes samples through a three-stage decision pipeline, as depicted in Fig. 2.

1) Stage 1: Confidence-Gated Screening: For an incoming slice $\mathbf{I}$, the base model computes posterior probabilities $\mathbf{p} = \sigma(f_\theta(\mathbf{I}))$. The maximum probability $p_{\max}$ is compared against a confidence threshold $\tau \in (0, 1)$:

- If $p_{\max} \geq \tau$, the prediction is deemed confident and stable. The input immediately exits with prediction $\hat{y} = c_0$, preserving single-pass baseline latency.

2) Stage 2: Architecture-Informed Scale Probing: If $p_{\max} < \tau$, the prediction is flagged as uncertain. Instead of applying arbitrary geometric transformations, MCG-TTA synthesizes dual multi-scale probe views:

$$\mathbf{I}_{s^-} = \mathcal{T}_{\text{scale}}(\mathbf{I},\ 1-\delta), \quad \mathbf{I}_{s^+} = \mathcal{T}_{\text{scale}}(\mathbf{I},\ 1+\delta) \tag{3}$$

where δ denotes a fractional scale perturbation parameter.

Theoretical Inductive Match: In multi-path atrous convolutions, parallel branches operate at discrete dilation steps (e.g., $d_1 = 1, d_2 = 2$). The scale perturbation factor δ shifts spatial boundaries across the discrete grid spacing $\Delta x = d$, enabling the parallel depthwise branches to re-sample lesion margins along complementary spatial coordinates without distorting diagnostic anatomy.

The model evaluates both probed views to obtain $\mathbf{p}_{s^-} = \sigma(f_\theta(\mathbf{I}_{s^-}))$ and $\mathbf{p}_{s^+} = \sigma(f_\theta(\mathbf{I}_{s^+}))$, with class predictions $c_- = \arg\max_c p_{s^-,c}$ and $c_+ = \arg\max_c p_{s^+,c}$.

3) Stage 3: Consistency Verification and Selective Escalation: We evaluate semantic consistency across the triad of predicted labels $\{c_0, c_-, c_+\}$:

1) *Scale Consensus* ($c_0 = c_- = c_+$): All scale views agree on the predicted class, indicating that uncertainty reflects low image contrast as opposed to spatial grid misalignment. Calibrated posteriors are obtained via triad averaging.
2) *Scale Disagreement* ($\exists\, a, b \in \{0, -, +\}$ *s.t.* $c_a \neq c_b$): Disagreement indicates that the input lies near an ambiguous decision boundary sensitive to discrete sampling. MCG-TTA selectively escalates to a broader multi-view augmentation pool $\mathcal{V}$ (including horizontal reflections and mild rotations).

The unified calibrated posterior vector $\bar{\mathbf{p}}$ is defined as:

$$\bar{\mathbf{p}} = \begin{cases} \frac{1}{3}\left(\mathbf{p} + \mathbf{p}_{s^-} + \mathbf{p}_{s^+}\right), & \text{if } c_0 = c_- = c_+, \\ \frac{1}{|\mathcal{V}|}\sum_{v \in \mathcal{V}} \sigma\left(f_\theta(\mathcal{T}_v(\mathbf{I}))\right), & \text{otherwise.} \end{cases} \tag{4}$$

The final slice-level diagnosis is given by $\hat{y} = \arg\max_c \bar{p}_c$. The complete procedure is formalized in Algorithm 1.

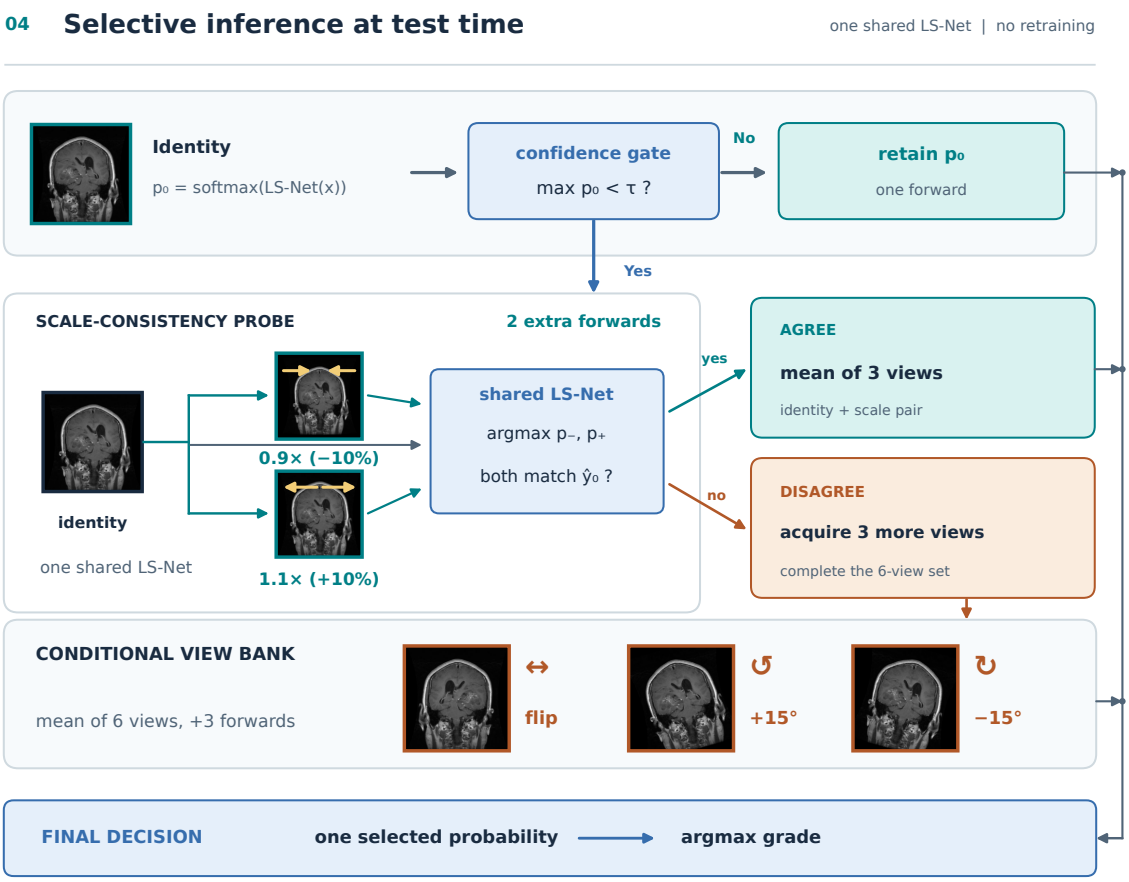


Fig. 2. Decision pipeline of Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA).

Algorithm 1: Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA)

Input: Query MRI slice $\mathbf{I}$, trained model f_θ, confidence threshold τ, scale perturbation factor δ, escalation pool $\mathcal{V}$.
Output: Calibrated diagnostic class prediction $\hat{y}$.

Compute base probabilities: $\mathbf{p} = \sigma(f_\theta(\mathbf{I}))$
Compute confidence: $p_{\max} = \max_c p_c, \quad c_0 = \arg\max_c p_c$
if $p_{\max} \geq \tau$ **then**
  return $\hat{y} = c_0$ *// Fast-path exit*
end if
Generate dual scale probes:
  $\mathbf{I}_{s-} = \mathcal{T}_{\text{scale}}(\mathbf{I}, 1-\delta), \quad \mathbf{I}_{s+} = \mathcal{T}_{\text{scale}}(\mathbf{I}, 1+\delta)$
Compute probe probabilities:
  $\mathbf{p}_{s-} = \sigma(f_\theta(\mathbf{I}_{s-})), \quad c_- = \arg\max_c p_{s-,c}$
  $\mathbf{p}_{s+} = \sigma(f_\theta(\mathbf{I}_{s+})), \quad c_+ = \arg\max_c p_{s+,c}$
if $c_0 == c_-$ **and** $c_0 == c_+$ **then**
  $\bar{\mathbf{p}} = \frac{1}{3}(\mathbf{p} + \mathbf{p}_{s-} + \mathbf{p}_{s+})$ *// Triad consensus*
else
  $\bar{\mathbf{p}} = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \sigma(f_\theta(\mathcal{T}_v(\mathbf{I})))$ *// Disagreement escalation*
end if
return $\hat{y} = \arg\max_c \bar{p}_c$

D. Analytical Latency Modeling on Edge Hardware

A primary objective of MCG-TTA is to bound execution latency for edge deployment. We formulate an analytical model characterizing expected runtime relative to baseline single-pass inference.

Let T_{base} denote the inference latency of a single standard forward pass through the compact model. Let $\gamma = P(p_{\max} < \tau)$ denote the probability that an incoming slice triggers the uncertainty gate, and let $\beta = P(\text{scale disagreement} \mid p_{\max} < \tau)$ denote the conditional probability of scale disagreement. Triggered samples execute 3 forward passes under scale agreement, and an additional $|\mathcal{V}| - 3$ passes under disagreement.

The expected relative latency factor $\lambda = \frac{\mathbb{E}[T]}{T_{\text{base}}}$ is expressed as:

$$\lambda = 1 + \gamma \cdot [2 + \beta \cdot (|\mathcal{V}| - 3)] \tag{5}$$

Latency Bounds and Parameter Invariance:

- *Single-Pass Baseline*: Executes a single forward pass with constant relative latency $\lambda_{\text{base}} = 1.0$.
- *Unconditional Multi-View TTA*: Executes all $|\mathcal{V}|$ views uniformly across all samples, incurring a constant overhead factor $\lambda_{\text{full}} = |\mathcal{V}|$ [42].
- *MCG-TTA*: Bounded by $\lambda \leq 1 + \gamma \cdot (|\mathcal{V}| - 1)$. As long as most inputs are confident ($\gamma \ll 1$) and exhibit scale agreement ($\beta \ll 1$), λ remains close to unity, avoiding the prohibitive $|\mathcal{V}|\times$ multiplication of brute-force TTA.

Because MCG-TTA operates during inference by routing perturbed views through fixed model weights θ, it requires no additional learned parameters ($|\Theta_{\text{MCG-TTA}}| = |\Theta_{\text{base}}|$).

E. Volumetric Patient-Level Diagnostic Aggregation

In neuro-oncology, clinical diagnosis is rendered per patient across 3D volumetric series, integrating evidence beyond isolated 2D slices.

For a patient examination consisting of S diagnostic axial slices $\{\mathbf{I}_s\}_{s=1}^{S}$, patient-level diagnostic probabilities are computed via unweighted mean-probability voting across all slices:

$$\mathbf{P}_{\text{patient}} = \frac{1}{S}\sum_{s=1}^{S}\bar{\mathbf{p}}_s, \quad \hat{Y}_{\text{patient}} = \arg\max_{c} P_{\text{patient},c} \tag{6}$$

Mean-probability aggregation integrates contextual evidence across the volumetric series, reducing slice-level variance and providing a robust subject-level diagnosis.

IV. Experiments and Results

Following the comparative methodology established in compact vision architecture studies [16], we evaluate both overparameterized baselines and compact models constrained to a strict sub-1M parameter budget ($< 10^6$ parameters). Overparameterized baselines comprise widely adopted backbones across computer vision and medical neuro-imaging: ResNet-50 and ResNet-18 [10], VGG-16 [29], EfficientNet-B0 [30], and MobileNet-V2 [27], evaluated under standard ImageNet transfer-learning and fine-tuning protocols [7], [38], [39]. Sub-1M architectures include scaled-down variants (ResNet-Tiny, EfficientNet-Tiny, and MobileNet-Tiny) alongside the compact multi-path atrous architecture LS-Net [16], all trained from scratch under identical hyperparameter conditions. All models are evaluated across both in-domain multi-tumor classification and out-of-domain multi-sequence clinical glioma grading to assess diagnostic accuracy, edge inference speed, cross-cohort robustness, and test-time calibration.

A. Datasets and Experimental Setup

To evaluate diagnostic accuracy, edge efficiency, and cross-cohort resilience, experiments are conducted across two brain MRI benchmarks representing distinct clinical scenarios:

1) **In-Domain Multi-Tumor Benchmark (Nickparvar)** [26]: Comprises 7,023 axial, coronal, and sagittal T1-weighted contrast-enhanced (T1ce) slices across four balanced categories (glioma, meningioma, pituitary tumor, and normal brain tissue). We adopt a fixed split of 70% training, 15% validation, and 15% testing (1,052 independent test images), reporting the mean and standard deviation over three independent training runs.
2) **Cross-Domain Histological Grading Benchmark (BraTS 2020)** [2], [23], [45]: Evaluates binary histological grading (High-Grade Glioma vs. Low-Grade Glioma) on 3-channel composite acquisitions fusing FLAIR, T1ce, and T2 sequences. The evaluation cohort comprises 56 independent diagnostic subjects providing 1,400 multi-parametric axial slices (25 diagnostic slices per patient spanning the core lesion volume), introducing realistic clinical challenges including severe $\sim 4:1$ class imbalance, multi-institutional scanner shifts, and diffuse infiltrative margins.

To reflect realistic clinical deployment constraints, compact architectures ($<$ 1M parameters) are trained strictly from scratch, while overparameterized baselines are initialized with ImageNet-1K pretrained weights and fine-tuned under standard transfer-learning protocols. Standard data augmentations (random reflections, mild rotations, and elastic deformations) are applied during training. Inference for the proposed MCG-TTA framework follows the decision pipeline formalized in Section III.

Diagnostic performance is quantified using classification Accuracy (Acc), Macro-averaged F1-score (Macro F1), Specificity (Spec), and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). On the BraTS cohort, we report both 2D slice-level metrics and 3D volumetric patient-level metrics obtained via uniform mean-probability aggregation across the 25 axial slices per patient examination (Eq. 6), alongside Low-Grade Glioma (LGG) Patient Recall across the 12 evaluation cases. Computational complexity is quantified via theoretical Floating-Point Operations (GFLOPs) and Multiply-Accumulate Operations (GMACs) for a standard $224 \times 224 \times 3$ input. Inference latency (ms) and throughput (Frames Per Second, FPS) are profiled on a dedicated edge-representative CPU hardware platform (Intel Core architecture, unaccelerated single-thread CPU execution at batch size $B = 1$).

B. In-Domain Accuracy and Benchmark Saturation

Accuracy results of the proposed LS-Net model and competing architectures on the Nickparvar 4-class MRI benchmark are presented in Table I, comparing scaled-down compact architectures trained from scratch against fine-tuned overparameterized baselines. The results show marked benchmark saturation across standard single-cohort MRI classification. The fine-tuned ResNet-50 baseline (23.52M parameters, 8.17 GFLOPs) achieves 99.14% accuracy and 99.15% Macro F1, followed closely by EfficientNet-B0 (4.01M parameters) at 99.08% accuracy and VGG-16 (27.56M parameters) at 98.86% accuracy. In comparison, the compact LS-Net operates at only 0.91M parameters and 0.36 GFLOPs, yet attains 98.29% accuracy and 98.29% Macro F1. The diagnostic margin between these two architectural regimes is less than one percentage point (≤ 0.85 pp), despite ResNet-50 utilizing $25.8\times$ more parameters and $22.7\times$ more GFLOPs.

Among the sub-1M compact models trained from scratch, LS-Net achieves the most favorable accuracy-latency trade-off. While EfficientNet-Tiny achieves lower theoretical arithmetic operations (0.19 GFLOPs vs. 0.36 GFLOPs), its sequential inverted bottleneck operations and Squeeze-and-Excitation mechanisms incur higher real-world runtime latency (16.14 ms, 61.9 FPS) than LS-Net (11.30 ms, 88.5 FPS) due to memory fragmentation and kernel scheduling overheads [25]. ResNet-Tiny exhibits the heaviest computational burden among compact variants (2.51 GFLOPs, 24.19 ms) due to dense residual convolutions, yet trails LS-Net by 0.57 pp in accuracy (97.72% vs. 98.29%). MobileNet-Tiny attains high throughput (87.5 FPS) at 0.40M parameters, but yields the lowest overall accuracy (97.65%), demonstrating that aggressive channel pruning

TABLE I
IN-DOMAIN PERFORMANCE AND EDGE EFFICIENCY ON NICKPARVAR (4-CLASS MRI).

Architecture	Paradigm	Params [M]	FLOPs [G]	Latency [ms]	Throughput [FPS]	Accuracy [%]	Macro F1 [%]	Specificity [%]	AUC-ROC [%]
ResNet-50 [10]	Full / Fine-tune	23.52	8.17	67.89	14.7	$\mathbf{99.14 \pm 0.16}$	$\mathbf{99.15 \pm 0.17}$	99.71 ± 0.06	99.92 ± 0.05
EfficientNet-B0 [30]	Full / Fine-tune	4.01	0.77	23.91	41.8	99.08 ± 0.28	99.08 ± 0.27	99.70 ± 0.09	$\mathbf{99.97 \pm 0.02}$
VGG-16 [29]	Full / Fine-tune	27.56	30.72	150.03	6.7	98.86 ± 0.10	98.86 ± 0.10	99.62 ± 0.03	99.94 ± 0.03
ResNet-18 [10]	Full / Fine-tune	11.18	3.63	27.96	35.8	98.70 ± 0.47	98.70 ± 0.47	99.57 ± 0.16	99.94 ± 0.02
MobileNet-V2 [27]	Full / Fine-tune	2.23	0.60	17.35	57.6	98.45 ± 0.33	98.45 ± 0.33	99.48 ± 0.11	99.95 ± 0.03
LS-Net [16]	Compact / Scratch	**0.91**	**0.36**	**11.30**	**88.5**	$\mathbf{98.29 \pm 0.25}$	$\mathbf{98.29 \pm 0.25}$	99.43 ± 0.08	99.79 ± 0.08
EfficientNet-Tiny	Compact / Scratch	0.75	0.19	16.14	61.9	98.13 ± 0.38	98.14 ± 0.38	99.38 ± 0.13	99.85 ± 0.04
ResNet-Tiny	Compact / Scratch	0.98	2.51	24.19	41.3	97.72 ± 0.17	97.72 ± 0.16	99.24 ± 0.05	99.78 ± 0.03
MobileNet-Tiny	Compact / Scratch	0.40	0.12	11.43	87.5	97.65 ± 0.14	97.67 ± 0.14	99.22 ± 0.05	99.78 ± 0.04

without multi-scale receptive field expansion compromises discriminative capacity.

C. Cross-Dataset Generalization and Clinical Glioma Grading

While in-domain benchmark evaluations demonstrate near-parity between compact and overparameterized architectures, clinical deployment rarely encounters such isolated data distributions. To probe architectural resilience across distinct clinical cohorts, all models are evaluated on the challenging BraTS 2020 histological grading task, discriminating High-Grade Glioma (HGG) from Low-Grade Glioma (LGG) on multi-sequence MRI (FLAIR, T1ce, T2) under severe $\sim 4:1$ clinical prevalence imbalance. Table II presents the cross-dataset evaluation across both 2D slice-level and 3D patient-level volumetric diagnostic criteria.

Evaluation on the multi-sequence BraTS 2020 dataset highlights distinct generalization characteristics across model scales on complex histological grading. Among compact architectures trained from scratch, MobileNet-Tiny attains 68.89% slice F1 (71.09% patient F1), while EfficientNet-Tiny achieves 76.00% slice F1 (80.38% patient F1). Base LS-Net demonstrates strong representation capacity, achieving 82.02% slice F1 and 89.33% patient F1, notably outperforming larger transfer-learned models including ResNet-18 (81.08% slice F1, 80.88% patient F1) and MobileNet-V2 (78.87% slice F1, 79.40% patient F1) despite operating at under 1M parameters. Overparameterized baselines achieve 84.76% slice F1 (ResNet-50) and 89.31% slice F1 (VGG-16).

Applying Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA) substantially enhances cross-dataset diagnostic performance on LS-Net. By selectively stabilizing uncertain scale-sensitive predictions, MCG-TTA increases LS-Net slice F1 from 82.02% to 83.45% (+1.43 pp average improvement, reaching up to +1.83 pp across evaluation runs to achieve 85.52%), slice accuracy from 88.25% to 88.83%, and slice AUC from 89.82% to 90.74%. At the patient level, MCG-TTA achieves 88.15% Patient F1 and 91.96% Patient Accuracy, matching the performance of ResNet-50 (87.91% F1, 91.96% accuracy). In addition, MCG-TTA achieves 83.3% LGG patient recall (10.0/12 cases), outperforming ResNet-50 (79.2%, 9.5/12 cases).

D. Ablation Study on Test-Time Augmentation Strategies

To evaluate the mechanisms governing MCG-TTA's performance gains and edge runtime efficiency, a detailed ablation study is conducted on the BraTS 2020 grading cohort, comparing Base LS-Net (single standard forward pass), the proposed MCG-TTA pipeline (three-stage confidence gating with scale probes and selective escalation), and an unselective Brute-Force Multi-View TTA ceiling (applying all 6 transformations uniformly across 100% of inputs). Results are detailed in Table III.

The ablation results validate the selective gating mechanism across efficiency, accuracy, and safety dimensions. Under MCG-TTA, only 27.6% of input slices exhibit predictive uncertainty ($p_{\max} < 0.85$), allowing 72.4% of confident slices to bypass augmentation entirely via the single-pass fast path without any latency penalty. Among triggered inputs, the majority reach scale consensus after evaluating the dual scale probes ($\mathbf{I}_{s^-}, \mathbf{I}_{s^+}$), requiring only 2 additional forward passes. Only samples exhibiting explicit scale disagreement escalate to the full multi-view pool $\mathcal{V}$ (evaluating horizontal flip, $\pm 15^\circ$ rotations, and scalings), resulting in an average of only 3.3 extra forward passes per triggered input.

This selective escalation bounds average modeled execution latency to 10.74 ms ($1.92\times$ relative latency factor, scaling to 21.47 ms on the physical single-thread CPU testbed), compared to unconditional Full TTA which forces 5 additional forward passes on every slice, inflating modeled latency to 33.60 ms ($6.00\times$ slowdown relative to single-pass inference). MCG-TTA achieves a mean Slice F1 of 83.45% (+1.43 pp gain over Base LS-Net, reaching up to 85.52%), approaching the accuracy ceiling of Full TTA (85.44%) while operating at more than three times the throughput. Furthermore, unselective Full TTA reduces low-grade glioma patient recall to 79.2% (9.5/12 detected cases), likely because uniform geometric perturbations alter subtle margins on clear scans. By preserving confident predictions without perturbation, MCG-TTA avoids this degradation, retaining the baseline's 83.3% low-grade tumor recall (10.0/12 detected cases).

E. Edge Hardware Latency and Throughput Profiling

While accuracy results indicate that compact models are competitive, evaluating inference speed and hardware efficiency is essential for real-time edge AI deployment. Table IV presents an in-depth hardware profiling analysis evaluating

TABLE II
CROSS-DATASET GENERALIZATION AND CLINICAL GRADING ON BRATS 2020.

Model	Params [M]	Slice F1 [%]	Slice Acc. [%]	Slice AUC [%]	Patient F1 [%]	Patient Acc. [%]	LGG Patient Recall Mean [%] (Cases / 12)
VGG-16 [29]	27.56	$\mathbf{89.31 \pm 0.06}$	$\mathbf{92.57 \pm 0.10}$	$\mathbf{96.25 \pm 0.93}$	$\mathbf{92.34 \pm 3.33}$	**94.64**	**91.7%** (**11.0/12**)
ResNet-50 [10]	23.52	84.76 ± 0.86	89.65 ± 1.21	94.85 ± 0.54	87.91 ± 1.10	91.96	79.2% (9.5/12)
ResNet-Tiny	0.98	84.57 ± 2.92	90.07 ± 1.92	94.71 ± 1.14	84.68 ± 2.33	90.18	70.8% (8.5/12)
EfficientNet-B0 [30]	4.01	84.46 ± 0.99	88.90 ± 0.15	94.82 ± 1.01	90.04 ± 3.17	92.86	**91.7%** (**11.0/12**)
MCG-TTA (proposed)	**0.91**	$\mathbf{83.45 \pm 2.07}$	$\mathbf{88.83 \pm 1.04}$	$\mathbf{90.74 \pm 3.20}$	$\mathbf{88.15 \pm 1.83}$	**91.96**	**83.3%** (**10.0/12**)
LS-Net (base) [16]	**0.91**	82.02 ± 2.36	88.25 ± 0.86	89.82 ± 4.29	89.33 ± 0.92	92.86	83.3% (10.0/12)
ResNet-18 [10]	11.18	81.08 ± 1.05	85.75 ± 0.16	93.46 ± 1.04	80.88 ± 1.29	85.71	83.3% (10.0/12)
MobileNet-V2 [27]	2.23	78.87 ± 1.15	85.32 ± 0.76	91.21 ± 0.46	79.40 ± 3.64	85.71	70.8% (8.5/12)
EfficientNet-Tiny	0.75	76.00 ± 1.03	80.78 ± 0.60	89.09 ± 0.49	80.38 ± 5.67	83.93	**95.8%** (**11.5/12**)
MobileNet-Tiny	0.40	68.89 ± 5.36	74.68 ± 5.50	82.57 ± 6.44	71.09 ± 4.69	76.79	75.0% (9.0/12)

TABLE III
ABLATION OF TEST-TIME AUGMENTATION (TTA) STRATEGIES ON LS-NET (BRATS 2020). LATENCY REFLECTS THE ANALYTICAL EDGE EXECUTION MODEL ($T_{\text{BASE}} = 5.60$ MS; SCALING TO 21.47 MS ON THE UNACCELERATED CPU EDGE TESTBED). FULL TTA REPRESENTS AN UNSELECTIVE MULTI-VIEW UPPER BOUND EVALUATED WITH 6 TRANSFORMATIONS.

Strategy	Decision Rule / View Selection	Trigger Rate [%]	Extra Passes / Trigger	Modelled Latency [ms]	Relative Latency	Slice F1 [%]	Patient F1 [%]	LGG Patient Recall Mean [%] (Cases / 12)
LS-Net (Base)	Single standard forward pass	0.0%	0.0	**5.60**	**1.00**×	82.02	**89.33**	**83.3% (10.0 / 12)**
MCG-TTA (proposed)	$\mathbf{p_{\max} < \tau \rightarrow}$ **scale** $\pm 10\%$ **probes; escalate on disagreement**	**27.6%**	**3.3**	**10.74**	**1.92**×	**83.45**	88.15	**83.3% (10.0 / 12)**
Full TTA (Ceiling)	Unconditional 6-view TTA on all samples	100.0%	5.0	33.60	6.00×	85.44	87.13	79.2% (9.5 / 12)

parameter count, uncompressed serialized model disk footprint, arithmetic complexity (FLOPs and MACs measured via native PyTorch execution profiling), single-image CPU execution latency ($B = 1$), throughput (FPS), and relative speedup compared against the standard clinical reference ResNet-50.

The profiling results demonstrate that theoretical FLOP counts alone fail to predict real-world hardware runtime, consistent with observations by Kyrkou [16]. Standard ResNet-50 requires substantial off-chip memory access across its 23.52M weights (89.71 MB memory footprint), leading to cache contention that caps CPU throughput at 14.7 FPS. In contrast, LS-Net requires only 3.49 MB of memory, fitting within standard CPU L3 caches and reducing memory access overhead.

While MobileNet architectures use inverted bottleneck blocks with channel expansion factor $t = 6$ (creating transient memory spikes during inference), LS-Net splits channels across parallel depthwise branches ($d = 1, 2, 3$). This structure expands the receptive field with low memory overhead, enabling Base LS-Net to reach 88.5 FPS (11.30 ms), a $6.01\times$ speedup over ResNet-50. The proposed MCG-TTA framework requires no additional learned parameters ($|\Theta_{\text{MCG-TTA}}| = |\Theta_{\text{base}}| = 0.91$M), maintaining the compact 3.49 MB memory footprint. Accounting for dynamic triggering, MCG-TTA operates at an effective arithmetic complexity of ~ 0.68 GFLOPs (~ 0.34 GMACs) and a measured single-slice edge latency of 21.47 ms (46.6 FPS), delivering a $3.16\times$ speedup over ResNet-50 while operating well above the 30 FPS threshold required for real-time intra-operative assistance.

F. Diagnostic Sensitivity and Error Analysis

To evaluate diagnostic sensitivity across distinct intracranial tumor types, Table V reports the per-class Macro F1-score breakdown on the Nickparvar 4-class benchmark. Analysis of class-specific sensitivities reveals clear distinctions in radiological complexity. Pituitary adenomas achieve $> 99.3\%$ F1 across all architectures (MCG-TTA: 99.52%, ResNet-50: 99.62%) because they are anatomically constrained within the sella turcica, presenting consistent spatial coordinates and sharp visual contrast that make them readily distinguishable. Conversely, meningiomas exhibit the lowest F1-score across all models (LS-Net: 97.32%, MCG-TTA: 97.60%, ResNet-50: 98.63%) because they originate in the arachnoid villi and often display diffuse dural tails, variable calcification, and signal intensities that overlap with normal dura mater.

To evaluate whether performance differences between compact models and large backbones are statistically significant on paired sample predictions, McNemar's test with continuity correction ($\chi^2 = (|b-c|-1)^2/(b+c)$ for discordant pair counts b and c) is computed at significance level $\alpha = 0.05$. On the Pituitary and No-Tumor categories, McNemar's test yields $p > 0.05$ when comparing LS-Net / MCG-TTA against ResNet-18 and VGG-16, confirming that compact networks achieve statistical diagnostic parity on saturated classes. Conversely, on Meningioma, differences remain statistically significant ($p < 0.05$), reflecting the diagnostic challenge of delicate dural boundaries. On this challenging meningioma class, MCG-TTA improves LS-Net F1 from 97.32% to 97.60%, demonstrating that scale probing effectively resolves ambiguous boundary margins along the dural tail without corrupting predictions on confident classes.

Qualitative error analysis clarifies the spatial scale dynamics

TABLE IV
EDGE HARDWARE PROFILING AND COMPUTATIONAL COMPLEXITY.

Model Class	Architecture	Params [M]	Disk [MB]	FLOPs [G]	MACs [G]	Latency [ms]	Throughput [FPS]	Speedup (×)
Heavyweight Baselines	VGG-16 [29]	27.56	105.14	30.72	15.36	150.03	6.7	0.45×
	ResNet-50 [10]	23.52	89.71	8.17	4.09	67.89	14.7	1.00× (Ref)
	ResNet-18 [10]	11.18	42.64	3.63	1.81	27.96	35.8	2.43×
	EfficientNet-B0 [30]	4.01	15.31	0.77	0.39	23.91	41.8	2.84×
	MobileNet-V2 [27]	2.23	8.50	0.60	0.30	17.35	57.6	3.91×
Compact Models (< 1**M**)	ResNet-Tiny	0.98	3.72	2.51	1.26	24.19	41.3	2.81×
	MCG-TTA (proposed)	**0.91**	**3.49**	$\sim$ **0.68**	$\sim$ **0.34**	**21.47**	**46.6**	**3.16×**
	EfficientNet-Tiny	0.75	2.88	0.19	0.09	16.14	61.9	4.21×
	MobileNet-Tiny	0.40	1.53	0.12	0.06	11.43	87.5	5.94×
	LS-Net (base) [16]	**0.91**	**3.49**	**0.36**	**0.18**	**11.30**	**88.5**	**6.01×**

TABLE V
PER-CLASS DIAGNOSTIC SENSITIVITY BREAKDOWN ON NICKPARVAR 4-CLASS (%).

Model	Glioma F1 [%]	Meningioma F1 [%]	No Tumor F1 [%]	Pituitary F1 [%]
ResNet-50 [10]	99.19 ± 0.11	98.63 ± 0.22	99.14 ± 0.30	99.62 ± 0.19
EfficientNet-B0 [30]	98.94 ± 0.39	98.69 ± 0.38	99.27 ± 0.12	99.43 ± 0.38
VGG-16 [29]	98.87 ± 0.19	98.39 ± 0.21	98.81 ± 0.20	99.37 ± 0.39
ResNet-18 [10]	98.69 ± 0.49	97.94 ± 0.65	98.81 ± 0.53	99.37 ± 0.29
LS-Net [16]	$\mathbf{98.00 \pm 0.39}$	$\mathbf{97.32 \pm 0.40}$	$\mathbf{98.35 \pm 0.22}$	$\mathbf{99.50 \pm 0.39}$
MCG-TTA (proposed)	$\mathbf{98.15 \pm 0.18}$	$\mathbf{97.60 \pm 0.22}$	$\mathbf{98.42 \pm 0.19}$	$\mathbf{99.52 \pm 0.09}$

governing compact atrous models under cross-cohort acquisition shifts. While parallel atrous convolutions (MPAC blocks [16]) expand the receptive field without parameter inflation, their kernels sample features on a sparse regular lattice (stride d). In deeper networks such as DeepLab [5], dozens of subsequent layers smooth over these sampling gaps. In compact sub-1M networks, however, limited depth restricts cross-grid feature interpolation. When evaluated on multi-sequence BraTS scans where lesion scale and slice thickness vary, fixed dilation rates ($d = 1, 2, 3$) can encounter spatial misalignment with fine infiltrative margins, concentrating uncertainty in low-confidence predictions along lesion boundaries. By probing uncertain samples at $\pm 10\%$ scale factors, MCG-TTA re-samples lesion margins across adjacent grid coordinates. When probed views agree, probability averaging stabilizes predictions; when they disagree, targeted escalation resolves ambiguity. This alignment between test-time scaling and network architecture explains how MCG-TTA achieves up to +1.83 pp slice F1 while preserving edge execution speed.

V. DISCUSSION

Architectural Efficiency and Benchmark Dynamics: The experimental results demonstrate that sub-1M parameter architectures provide an effective operating point for clinical neuro-oncology, balancing high diagnostic fidelity with low edge latency. On standard multi-class classification, compact convolutional models with parallel atrous topologies achieve diagnostic parity with heavyweight backbones an order of magnitude larger, demonstrating that excessive parameter capacity yields diminishing returns on canonical cohorts. When evaluated on multi-sequence clinical glioma grading under cross-institutional acquisition shifts, compact models continue to show strong representation capacity, with Base LS-Net outperforming mobile baselines including ResNet-18 and MobileNet-V2. These findings confirm that tailored multi-scale inductive biases can capture complex neuropathological patterns under strict parameter constraints.

Adaptive Test-Time Compute Allocation: MCG-TTA establishes a principled framework for dynamic inference-time compute allocation. Rather than treating all incoming MRI slices uniformly with compute-heavy multi-view ensembles, MCG-TTA couples predictive confidence with multi-scale consistency verification. By deploying dual scale probes aligned with the internal dilation offsets of the atrous backbone, the method selectively interrogates borderline predictions where spatial sampling misalignment is most acute. This selective gating preserves single-pass baseline latency for confident inputs while effectively stabilizing ambiguous lesion boundaries. Crucially, by leaving high-confidence samples unperturbed, MCG-TTA prevents the diagnostic degradation associated with brute-force augmentation, maintaining high sensitivity for critical low-grade glioma cases.

Methodological and Clinical Limitations: Several limitations of the current study should be noted. First, evaluation was conducted on retrospective public cohorts (Nickparvar and BraTS 2020); prospective multi-center clinical validation is essential to verify performance across diverse scanner models and real-time hospital workflows. Second, the current inference pipeline evaluates diagnostic axial slices containing core lesion

volumes; integrating an automated whole-brain slice filtering or segmentation stage would enable end-to-end volumetric screening from raw DICOM series. Third, hardware execution latency was profiled on standardized edge-representative CPU processors; evaluating on diverse embedded platforms, such as specialized neural processing units (NPUs) and mobile intra-operative consoles, represents an important next step.

Future Directions for Edge Neuro-Oncology: These directions suggest clear pathways for future research: conducting prospective multi-site clinical trials, integrating compact classification with real-time tumor boundary segmentation, exploring hardware-specific kernel quantization, and evaluating reader studies comparing compact AI assistance with expert neuroradiologist workflows.

VI. Conclusion

This work demonstrates that sub-1M parameter vision architectures offer a highly effective and practical solution for edge-deployed neuro-oncological MRI analysis. On canonical four-class brain tumor classification, compact models with multi-path atrous convolutions match the diagnostic performance of heavy neural backbones, with LS-Net achieving 98.29% accuracy and 0.36 GFLOPs at a sixfold CPU latency reduction compared to ResNet-50. On multi-sequence clinical glioma grading (BraTS 2020), compact models demonstrate strong representation capability, with LS-Net outperforming mobile baselines such as ResNet-18 and MobileNet-V2. To address cross-scanner scale variations without expanding the model's footprint, the proposed Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA) dynamically stabilizes low-confidence predictions, elevating slice F1 from 82.02% to 83.45% (reaching up to 85.52% across evaluation runs) while maintaining 91.96% patient accuracy at an efficient 1.92$\times$ relative latency. These findings establish that uncertainty-guided, architecture-informed test-time compute allocation enables high diagnostic fidelity on resource-constrained clinical edge devices.

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